



**MAKERERE
UNIVERSITY**
"We Build For The Future"

Week 1

BSE3214

Introduction to Cloud Computing Part 1

| Lecture Material



Instructor

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Overview

Lecture Agenda & Key Topics



What is Cloud Computing?



Cloud Service Models: SaaS, PaaS, IaaS



Historical Evolution



Cloud Deployment Models



Key Drivers of Cloud Computing



Security Considerations



Characteristics of Cloud Computing



Limitations and Challenges

What is Cloud Computing?

Core Concepts & Definition



Core Definition

"The delivery of **on-demand** network access to shared pools of **configurable** computing resources."

- NIST Definition (Simplified)

Resource Types

 Servers

 Storage

 Databases

 Networking

 Software

 Analytics



Over the Internet

Resources are provided remotely via "the cloud" rather than local hardware.



On-Demand Access

Users can provision resources automatically without human intervention.

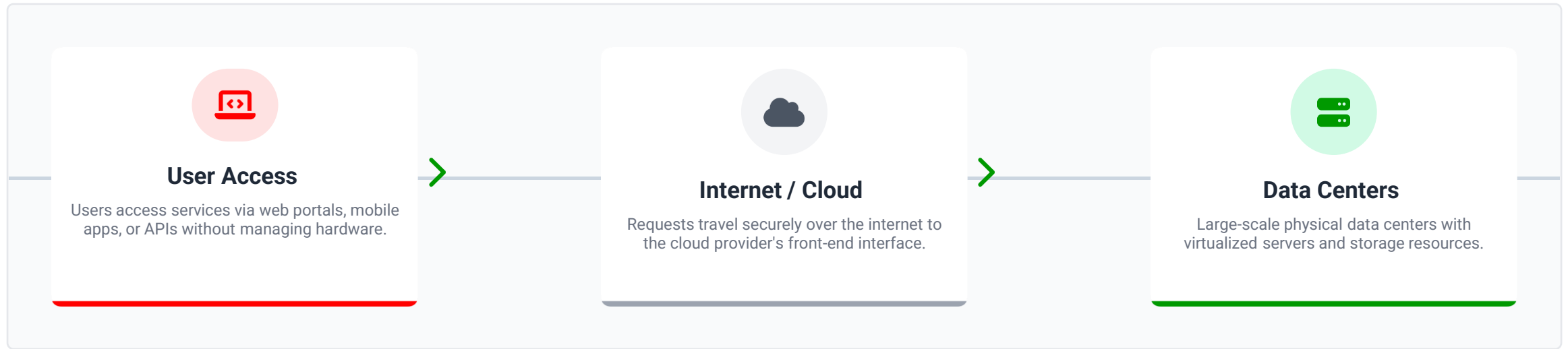


Pay-Per-Use Model

Utility-based consumption similar to electricity or water billing.

How Cloud Computing Works

Architecture & Mechanism



Virtualization Layer

Physical hardware is abstracted into virtual machines (VMs), allowing multiple users to share resources securely and efficiently.



Seamless Elasticity

The infrastructure scales up or down instantly to handle workload spikes, ensuring consistent performance.



Automated Provisioning

Orchestration software automatically allocates and de-allocates resources based on user demand without human intervention.



Abstraction of Complexity

Users focus on their applications and data while the cloud provider manages the underlying hardware, maintenance, and updates.

Historical Evolution

Timeline of Cloud Computing Development



1960s



Mainframes



Centralized Mainframes

Time-sharing of expensive computing resources among multiple users.

Centralized computing resources shared via dumb terminals.

Utility computing concept born

Early 2000s



Virtualization

Hardware abstraction allows efficient resource utilization (VMware, Xen).

1980s-90s



Client-Server

PCs connected to powerful servers.
Distributed computing emerges.

Mid 2000s



Cloud Platforms

Launch of AWS (2006), Azure, GCP.
IaaS, PaaS, SaaS models defined.

Key Drivers

Why Cloud Computing?



Scalability

Easily grow or shrink resources based on demand.



Elasticity

Automatic, real-time resource adjustment.



Cost Efficiency

Pay-per-use model reduces upfront capital innovation.



Flexibility

Wide variety of services and deployment options.



Global Reach

Deploy worldwide in minutes for low latency.

Five Essential Characteristics

NIST Definition of Cloud Computing



On-Demand Self-Service

Automatic provisioning of computing capabilities like server time and network storage without requiring human interaction.



Broad Network Access

Capabilities are available over the network and accessed through standard mechanisms (mobile, tablets, laptops, workstations).



Resource Pooling

Provider's computing resources are pooled to serve multiple consumers using a multi-tenant model with dynamic assignment.



Rapid Elasticity

Capabilities can be elastically provisioned and released to scale rapidly outward and inward commensurate with demand.



Measured Service

Resource usage is monitored, controlled, and reported, providing transparency for both the provider and consumer.

Source: NIST SP 800-145

Traditional IT vs Cloud Computing

Comparative Analysis of Infrastructure Models



Feature	Traditional IT	Cloud Computing
Ownership	 Own hardware & data center	 Provider owns infrastructure
Setup Time	 Weeks to months (procurement)	 Within minutes / hours
Cost Model	 High upfront cost (CapEx)	 Pay-as-you-go (OpEx)
Scalability	 Manual, physical upgrades	 Automated, elastic scaling
Maintenance	 Internal IT department manages all	 Shared responsibility model
Accessibility	 Mostly local / VPN restricted	 Global access via Internet



SaaS

Software as a Service

End Users



PaaS

Platform as a Service

Application Developers



IaaS

Infrastructure as a Service

IT Administrators / Network Architects

Physical Infrastructure (Data Centers)

1. Software as a Service

Software hosted and managed by providers, accessed via web browsers or mobile apps. No installation required.

Gmail

Salesforce

Office 365

2. Platform as a Service

Platform for application development and deployment. Includes OS, runtime, and middleware managed by provider.

Google App Engine

Heroku

Azure App Service

3. Infrastructure as a Service

Virtualized computing resources over the internet. Users control OS, storage, and deployed applications.

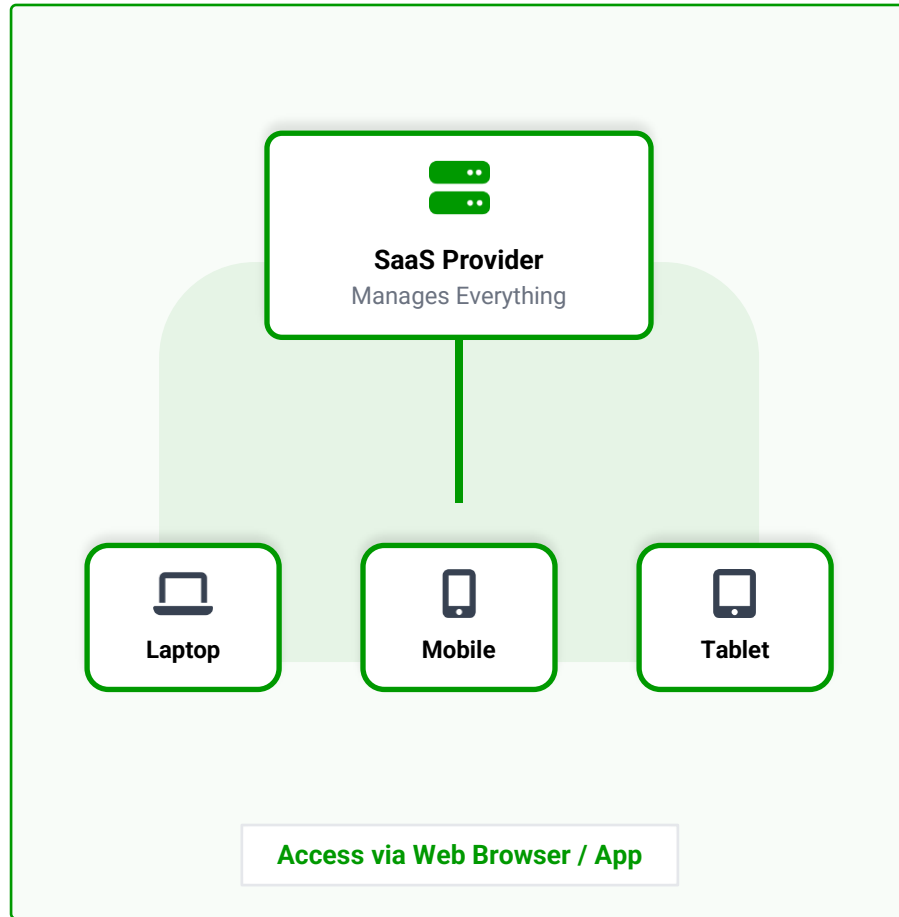
AWS EC2

Google Compute Engine

Azure VMs

SaaS - Software as a Service

Fully Managed Applications



Definition

Applications delivered online over the internet, fully managed by the provider. Users access software without installation.

Architecture

Multi-tenant applications hosted remotely. Single instance serves multiple customers (tenants) with data isolation.

Control & Benefits

User controls data & settings only. No infrastructure worries. Instant use, automatic updates, accessible anywhere.

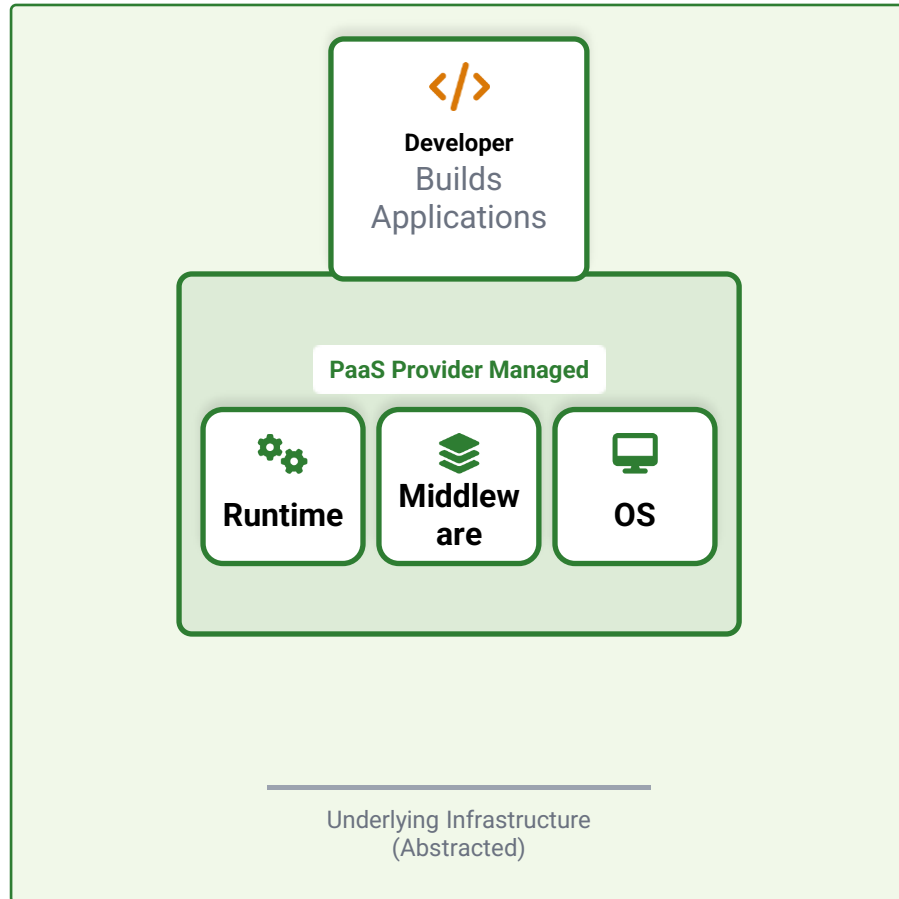
Limitations

Limited customization options, internet connectivity dependent, potential data privacy concerns regarding provider access.



PaaS - Platform as a Service

Development & Deployment Environment



i Definition

Cloud platform providing hardware and software tools (OS, middleware, runtime) over the internet. Developers build applications without managing infrastructure.

🔗 Architecture

Customers manage applications and data. Provider manages the underlying platform (runtime, middleware, OS, virtualization, servers, storage, networking).

✓ Control & Benefits

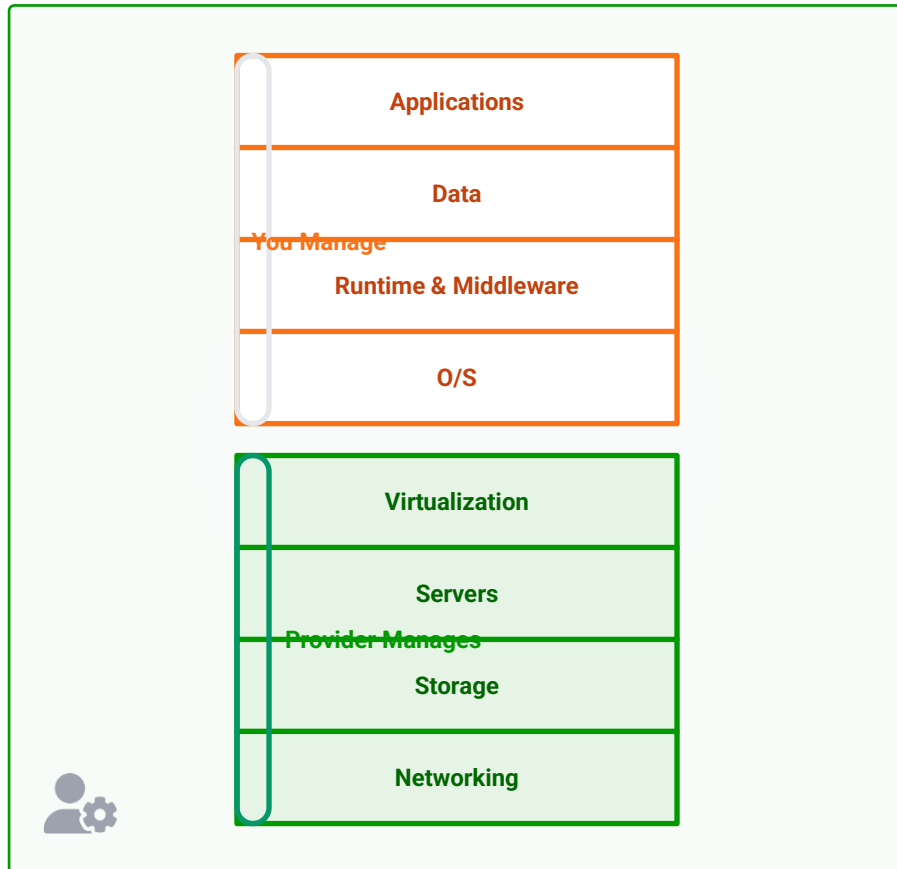
Faster application development and deployment. Reduces maintenance overhead. Automatic scaling and load balancing.

⚠️ Limitations

Vendor lock-in risk (proprietary tools), limited control over the underlying environment, potential compatibility issues.

IaaS - Infrastructure as a Service

Virtual Computing Resources & Control



POPULAR EXAMPLES



Definition

Fundamental building blocks for cloud computing. Provides virtualized servers, storage, and networking online with full control over the OS and applications.

Architecture

Provider manages the physical data centers and virtualization layer (hypervisor). Users are responsible for everything above: Guest OS, patches, and software.

Control & Benefits

Maximum flexibility and control. Ideal for "lift and shift" migrations, complex custom applications, and legacy system hosting.

Limitations

Higher operational overhead. User is responsible for OS updates, security patches, and middleware configuration (Security is a shared responsibility).



SaaS vs PaaS vs IaaS

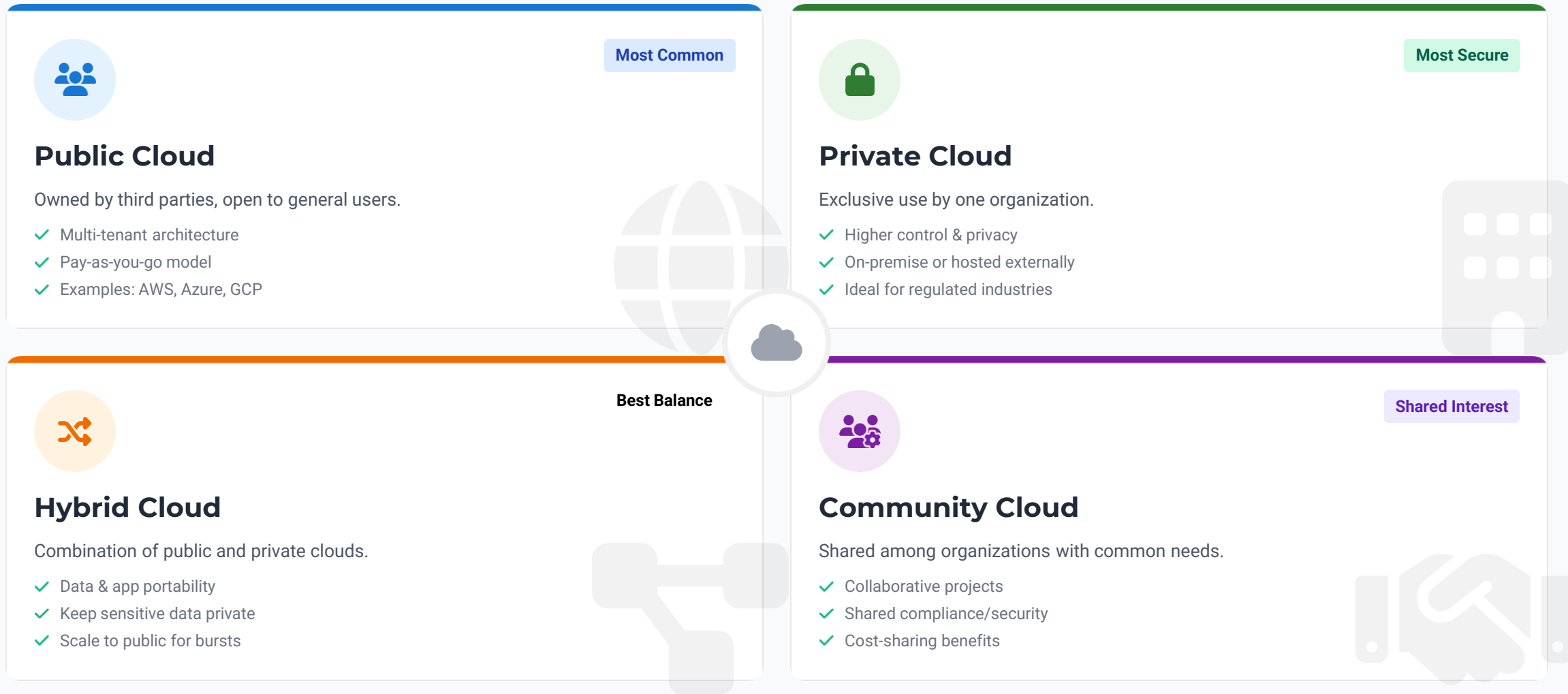
Service Model Comparison Matrix



	SaaS	PaaS	IaaS
User Control	App & Data Settings Minimal control	Applications + Data Moderate control	OS + Apps + Data High control
Provider Managed	Full Stack (except data) Everything managed	Platform + Infra Runtime & OS managed	Infra + Virtualization Hardware managed
Ideal For	End Users Business consumers	Developers Software engineers	IT Admins / Architects System administrators
Examples	Google Workspace Salesforce	App Engine Azure App Service	AWS EC2 Compute Engine
Complexity	Low / Easiest 	Medium 	High / Complex

Cloud Deployment Models

Four Main Ways to Deploy Cloud Services



Cloud Security Considerations

Critical Aspects for Safe Adoption



Multi-tenancy & Isolation Risks

Potential for data leakage between tenants if isolation mechanisms fail. "Noisy neighbor" issues affecting performance and availability.



Data Privacy & Encryption

Importance of encrypting data at rest (storage) and in transit (network). Ensuring provider doesn't access sensitive data without authorization.



Regulatory Compliance

Adhering to laws like GDPR, HIPAA, and local data sovereignty regulations. Ensuring cloud provider certifications match organizational needs.



Identity & Access Management (IAM)

Implementing strong authentication (MFA), least privilege access policies, and robust role-based access control (RBAC) to prevent unauthorized access.



Shared Responsibility Awareness

Understanding that security is a joint effort. Users must secure their data and applications, while providers secure the underlying infrastructure.

Shared Responsibility Model

Security & Compliance in the Cloud



Cloud Provider Responsibilities

"Security OF the Cloud"



Physical Infrastructure

Data centers, hardware, cabling, power & cooling



Physical Security

Guards, fences, biometric access, surveillance



Host Infrastructure

Hypervisor isolation, network segmentation



Storage Encryption

Infrastructure for encryption at rest



SaaS Application Security

Full stack security for SaaS offerings



Customer Responsibilities

"Security IN the Cloud"



Network Security

Firewall configuration, ACLs, VPC settings



OS & Patch Management

Guest OS updates, middleware patching



Identity & Access

User accounts, roles, MFA, permissions



Data Governance

Data classification, retention policies



Data Encryption

Client-side encryption, key management

Limitations & Challenges

Critical Considerations for Cloud Adoption



Network Dependency & Latency

Services are entirely dependent on internet connectivity. High latency can affect performance of real-time applications.



Risk of Downtime & Outages

Service disruptions can occur (e.g., historical AWS S3 outages). Business continuity plans are essential.



Vendor Lock-in

Proprietary formats and APIs can make it difficult and costly to switch providers or migrate data.



Legal, Regulatory & Data Sovereignty

Compliance with laws like GDPR involves strict rules on where data is stored and how it is processed.



Ongoing Security Threats

Shared infrastructure introduces risks. Misconfigurations and credential theft remain top attack vectors.

Summary

Key Takeaways & Conclusions



Cloud Transforms IT

Offers scalable, flexible, and cost-efficient solutions replacing traditional CapEx models.



Service Models Mastery

Deep understanding of SaaS, PaaS, and IaaS is crucial for architectural decisions.



Deployment Strategy

Choose between Public, Private, Hybrid, or Community clouds based on organizational needs.



Security is Shared

Security is a joint responsibility between the cloud provider and the customer.



Evaluate Limitations

Consider network dependency, latency, and compliance constraints before adoption.

References & Further Reading

Recommended Resources for Deepening Knowledge



NIST SP 800-145

The NIST Definition of Cloud Computing

Mell, P., & Grance, T. (2011). National Institute of Standards and Technology.



Cloud Security Alliance (CSA)

Industry standard research and best practices for cloud security.

Security Guidance for Critical Areas of Focus in Cloud Computing.



Official Cloud Documentation

Comprehensive guides and whitepapers from major providers.

[AWS Documentation](#)

[Azure Docs](#)

[Google Cloud Docs](#)



Research Articles

Academic perspectives on cloud economics and security challenges.

InfoQ, IEEE Xplore, and ACM Digital Library resources.



Students are encouraged to explore the **Makerere University Library** e-resources for access to full-text journals and conference proceedings related to cloud computing trends in East Africa.



Questions & Discussion

Interactive Session



Any Questions?

Feel free to ask about cloud concepts, service models, deployment strategies, or security implications discussed today.



Clarification on Service Models (SaaS vs PaaS vs IaaS)



Security Challenges & Shared Responsibility



Real-world Application Scenarios